

Evidence-grounded summaries: a synthetic consistency illustration

Abstract

A fluent summary can contain factual statements that its source does not support. This note describes a source-support check and illustrates a paired comparison with six synthetic scores. Mean illustrative consistency is 87% for the baseline and 91% for the evidence-grounded condition, a difference of four percentage points. The scores are authored demonstration values, not outcomes of a model evaluation or an annotation study.

Problem and mechanism

The research question is whether associating each factual statement with source spans can reduce unsupported content without losing important information. The candidate first drafts a summary, associates factual statements with supporting spans, and revises or omits statements lacking support. The proposed mechanism is the support check, so the appropriate control removes that check while holding the source, model, length budget and decoding settings constant.

Evaluation definition

Consistency is the fraction of factual summary statements supported by their source. A real annotation study would predefine segmentation, the denominator and how unverifiable statements are treated. Coverage and readability are necessary companion measurements because consistency alone can reward omission.

Demonstration results

data/results.csv has three paired illustrative seeds per condition. Each condition has a sample standard deviation of one percentage point.

Condition	Mean consistency (%)	Difference from baseline (percentage points)
Baseline	87	0
Evidence-grounded	91	4

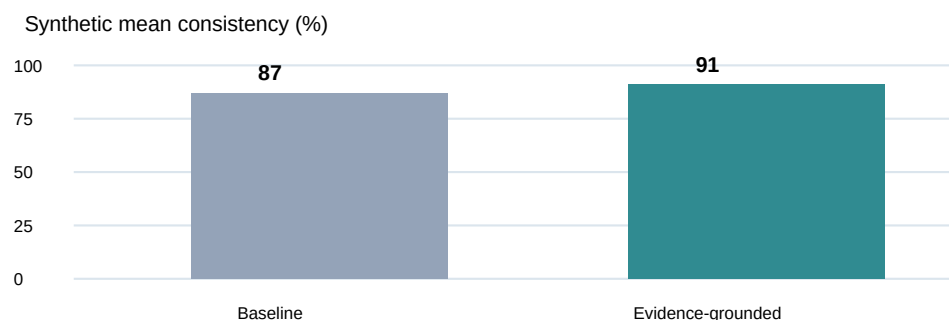
Each paired row differs by four percentage points. Figure 1 shows the means on a 0–100 scale. code/analyze.py reproduces the means and paired differences without third-party dependencies. Percentages and percentage-point changes are kept distinct.

Evidence boundary

No summaries were generated or annotated for this illustration. The synthetic rows contain no coverage, readability, human agreement, latency or token-cost evidence. No significance or generalization claim follows from the constant paired difference. A real mechanism test requires a support-check ablation with paired documents and blinded annotation.

Conclusion

The example makes the result table, arithmetic and research claim agree. Source grounding is a testable mechanism, but its factuality benefit and information-retention cost remain empirical questions.



基于证据的摘要：合成一致性示例

摘要

流畅的摘要也可能包含源文无法支持的事实性陈述。本札记介绍源文支持性检查，并以六个合成分数说明配对比较。普通方法的演示平均一致性为 87%，基于证据的条件为 91%，相差 4 个百分点。这些分数是编写的演示数值，不是模型评测或标注研究的结果。

问题与机制

研究问题是：为每条事实性陈述关联源文片段，能否减少不受支持的内容，同时保留重要信息？候选机制先形成摘要，再为事实性陈述关联支持片段，对缺少支持的陈述进行修订或省略。拟研究的机制是支持性检查，因此合适的对照是移除该检查，同时固定源文、模型、长度预算和解码设置。

评测定义

一致性定义为摘要中被源文支持的事实性陈述比例。真实标注研究需预先定义切分方式、分母及不可验证陈述的处理。覆盖率和可读性是必要的互补测量，因为只看一致性可能奖励遗漏。

演示结果

data/results.csv 每个条件包含三个配对演示种子，样本标准差均为 1 个百分点。

条件	平均一致性 (%)	相对普通方法的差异 (百分点)
普通方法	87	0
基于证据	91	4

每个配对行相差 4 个百分点。图 1 在 0 到 100 的量表上展示均值。code/analyze.py 不依赖第三方包，可复算均值和配对差异。百分比与百分点变化明确区分。

证据边界

本示例没有生成或标注摘要。合成行不包含覆盖率、可读性、人工一致性、延迟或 token 成本证据。不能从恒定的配对差异推导显著性或泛化结论。真实机制检验需要基于配对文档及盲法标注的支持性检查消融。

结论

本例使结果表、计算与研究结论一致。源文支持性检查是可检验的机制，但其事实性收益和信息保留代价仍需实证回答。

